

Project 05: Multi-Agent SDR Platform

Autonomous Sales Development Representative Assembly Line

1. Executive Overview

The Multi-Agent SDR Platform orchestrates a 4-agent collaborative pipeline for B2B sales engineering. It enriches raw leads, evaluates ICP fit scores, synthesizes persona-specific outreach hooks, drafts personalized cold emails, and enforces a mandatory Human Approval Gate before dispatch.

2. System Architecture & Workflow Pipeline

- Agent 1: Lead Discovery & Profile Enrichment
- Agent 2: ICP Qualification Scoring (0-100 Scale)
- Agent 3: Industry Pain Point & Persona Hook Synthesis
- Agent 4: Personalized Cold Email Copywriter
- Human Approval Safety Gate -> Outbound Sales Dispatch

3. API Endpoints Specification

Method	Endpoint	Description
GET	/	Serves Multi-Agent SDR Platform web UI
GET	/api/leads	Returns available B2B lead records
POST	/api/process-lead	Executes 4-agent SDR workflow for selected lead
POST	/api/approve-email	Human approval gate endpoint to finalize outreach email

4. Deployment & Automated Verification

Standalone Uvicorn Launch:

```
cd 05_multi_agent_sdr/src && python -m uvicorn server:app --host 127.0.0.1 --port 8005
```

Automated Test Suite Execution:

```
python -m pytest 05_multi_agent_sdr/tests/
```